

EvoSensory Beats

MANUAL · QUICK START & REFERENCE

Multi-sensory rhythmic timing training, built by a pediatric OT. This guide covers everything you need to run your first session, understand the data, and document it.

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THE EVIDENCE

Why Rhythmic Timing Training Works

Rhythmic timing training has a substantial evidence base built across more than three decades of research in neuroscience, occupational therapy, physical therapy, and special education. The brain's internal timing system — involving the cerebellum, basal ganglia, supplementary motor area, and prefrontal cortex — can be trained through structured, repetitive practice synchronizing movement to an external beat.

When this system is less efficient, the effects are visible in daily life: impulsivity, bilateral coordination difficulty, slow processing speed, reading challenges, handwriting struggles, and poor motor sequencing. EvoSensory measures timing errors to the millisecond and makes them documentable in a way a therapist watching cannot achieve with the naked eye.

EvoSensory Beats — the evidence-based core

EvoSensory Beats is the only module with direct grounding in the rhythmic timing training evidence base. It implements the specific mechanism — voluntary movement synchronized to an external auditory beat with immediate multimodal error feedback — that peer-reviewed literature has measured and validated.

The other four modules are

evidence-informed extensions

that apply timing principles to additional clinical domains. Use EvoSensory Beats as the primary intervention; add other modules when the target domain is clinically relevant.

Area	Research Finding	Key Reference
Sensorimotor synchronization	Core mechanism: voluntary movement can be trained to synchronize with external auditory beats via cerebellar circuits.	Repp & Su (2013). Psychonomic Bulletin & Review. PMID: 23397235
ADHD & Attention	Children with ADHD show measurable timing deficits independent of inattention. Rhythmic training reduces impulsivity through cerebellar-prefrontal pathways.	Zelaznik et al. (2002). Journal of Motor Behavior. PMID: 12521818
DCD & Bilateral Coordination	Rhythmic timing training improves alternating limb timing and interlimb synchronization in DCD, with functional transfer.	Zelaznik & Goffman (2010). Adv Exp Med & Biol. PMID: 19876484
Reading & Phonological Processing	Temporal processing underlies phoneme discrimination. Auditory-motor entrainment improves phonological skills and reading fluency.	Tierney & Kraus (2014). Front Human Neuroscience. PMID: 25538596
Basal Ganglia & Beat Timing	Beat-based timing is mediated by the basal ganglia and can be specifically trained through sensorimotor practice.	Grahn & Rowe (2009). Journal of Neuroscience. PMID: 19474308

Area	Research Finding	Key Reference
Motor Planning & Praxis	Sequenced rhythmic training enhances motor program formation and execution speed in praxis deficits.	Molinari et al. (2007). Cerebellum. PMID: 17687732
TBI & Acquired Injury	Post-TBI timing deficits correlate with processing speed and cognitive-motor outcomes. Rhythmic training shows measurable improvement.	Thaut et al. (2009). Neuropsychological Rehabilitation. PMID: 19205969
Parkinson's Disease	RAS is one of the most supported non-pharmacological PD interventions — documented improvements in gait cadence and motor timing.	Thaut et al. (2015). Curr Neurology & Neuroscience Reports. PMID: 26407995
Stroke Rehabilitation	Rhythmic cuing improves motor timing and bilateral retraining in post-stroke upper and lower extremity rehabilitation.	Thaut & Abiru (2010). Music Perception. DOI: 10.1525/mp.2010.27.4.263
Handwriting	Graphomotor timing is one of the strongest predictors of handwriting legibility and speed. Rhythmic practice produces transfer effects.	Longcamp et al. (2008). Neurolmage. PMID: 17855113

Why millisecond precision matters

8ms early versus 62ms early is invisible to the human eye but clinically significant. The 8ms patient has an established internal clock. The 62ms patient has a systematic anticipatory pattern that mirrors impulsivity in daily functioning. EvoSensory makes that distinction visible and documentable in every session.

STEP BY STEP

Quick Start

- 1 **Open the app and create a patient profile**
Go to www.evolve-ot.com/evosensory.html in Chrome. Tap + **New Player**. Enter name or initials, select age category, choose training modality. BPM is set automatically. Bookmark the page or tap Share → Add to Home Screen for easy access.
- 2 **Plug in wired headphones**
Left ear tone = tapping early. Right ear tone = tapping late. This is the core clinical mechanism — Bluetooth adds 100–200ms of lag that defeats it entirely. Any wired earbuds work. **Do this before every session.**
- 3 **Choose a module and level**
Select **EvoSensory Beats** → **Level 0 or 1**. EvoSensory Beats is the evidence-based core — use it with every new patient. Adjust BPM with + / –.
- 4 **Confirm hardware**
A setup screen appears before each session. Check what is connected — wired headphones, device vibration, XAC cushion. Only what you confirm is documented in the SOAP note.
- 5 **Tap with the beat**
Tap the large button, tap the canvas, or press **spacebar** on desktop. One tap per beat.
- 6 **Generate SOAP note**
When the session ends, fill in the Session Observation field, then tap **Generate SOAP Note**. Edit before pasting into your EMR.

Use wired headphones — required for spatial audio

The spatial audio feedback is the core clinical mechanism — early taps produce a tone in the left ear, late taps produce a tone in the right ear. Without wired headphones this feedback is not available. Bluetooth adds 100–200ms lag that defeats this entirely. A \$10 wired earbud is all you need.

SETUP

Patient Profiles

Category	Hands BPM	Feet BPM	Notes
 Early Childhood (3–7)	52 BPM	44 BPM	150ms window · Short attention, developing gross motor
 School Age (8–17)	60 BPM	52 BPM	120ms window · Standard pediatric baseline
Adult (18–64)	66 BPM	56 BPM	Slightly faster baseline
Older Adult (65+)	52 BPM	44 BPM	Wider hit window applied automatically

Feet start 8–10 BPM slower: hip → knee → ankle → floor is a longer motor sequence than a hand tap. On phones/tablets, verify bilateral hand use or select the appropriate single-hand modality.

THE FIVE MODULES

Choosing the Right Module

Module	What It Trains	BPM	Requires
 EvoSensory Beats	Core timing, bilateral coordination, spatial audio + haptic feedback. 8 levels (0, 1, 1b, 2–6). 4 visual games: Falling Notes, Hurdle Jump, Target Range, Money Catcher. Start every patient here.	Adjustable	Any device
 EvoSensory Vision	Evidence-informed. 5 sub-modes: Smooth Pursuits (tap), Smooth Pursuits 2 (camera gaze, no tap), Saccadic Jumps (tap on jump), Whack-a-Mole (reaction tap L/R), Attention Hold (stare without looking away — magic explosion reward).	Adjustable / Default: 44 BPM	Any device Pursuits 2: Chrome desktop
 Cognitive	Vocabulary: app announces word → GO! → letter beats → whole word. 2 loops per word. Track Accuracy toggle: off = listen-only. Math: spoken number sequences on the beat.	Adjustable	Any device
 Motion	Camera detects clapping, knee taps, marching, and jumps timed to the beat.	Adjustable	Chrome on computer
 Active Games	Body movement controls 5 games — catch stars, pop balloons, kick soccer, duck/jump, balance zones.	—	Chrome on computer

EvoSensory Vision default BPM is 44

Visual tracking needs more time per beat than hand tapping — the eyes must locate the target, track it, and reset before the next beat. Start at 44 BPM and increase gradually.

EvoSensory Beats — Level Progression

Level	Pattern	Advance When
0 —  First Beat	One side only, unilateral, 48 BPM. Best entry point for younger/ lower-function patients.	85%+ accuracy, 2 sessions
1 —  Warm Up	Single button, any key, every beat, 56 BPM	85%+ accuracy, 2 sessions
1b —  Both Hands	Both sides simultaneously, 56 BPM. Bilateral synchronization foundation.	85%+ accuracy, 2 sessions
2 —  Finding It	Alternating L/R, 66 BPM	85%+ accuracy, 2 sessions
3 — In the Groove	Alternating with rests	80%+ accuracy, stable trend
	Complex L/R sequences + rests	80%+ accuracy, <50ms deviation

Level	Pattern	Advance When
4 — Steady Beat		
5 — Beat Master	Irregular patterns + rests	80%+ accuracy, consistent timing
6 — Foot Tap	Bilateral foot patterns	Clinical ceiling — maintain for consolidation

BPM progression rule: Increase tempo by 4–8 BPM at a time. If accuracy falls below 50%, reduce BPM before changing the level.

Motion — Sub-Modes

Chrome on computer required

- Clapping (bilateral UE, waist-up)
- Knee Taps / Midline Crossing (waist-head)
- Marching (full body)
- Jumps (full body + clearance)
- Reach & Catch (trunk rotation, balance, visual-motor)

Active Games — Sub-Modes

Chrome on computer required

- Catch Stars (lateral stepping, vestibular)
- Pop Balloons (overhead reach)
- Kick Soccer (unilateral foot)
- Duck & Jump (motor planning)
- Side Steps & Balance (static/dynamic)

READING THE DATA

Understanding Your Results

1. Accuracy — Were they hitting the beat?

% of beats hit within the acceptable time window. Reflects engagement, attention, and task level. **High accuracy alone does not mean good timing.**

2. Timing Deviation — Where exactly did taps land?

Measured in milliseconds (ms). Lower = better. Zero = on the beat. **(+) = early (before the beat) (-) = late (after the beat)**

Accuracy and deviation are not the same thing
A patient can score 95% accuracy but have +55ms average deviation — hitting the window every time, but always early. That is a separate, clinically meaningful finding that accuracy alone does not capture.

3. Session Trend — Did timing change during the session?

Tracks whether errors grew, shrank, or held steady. Drifting earlier may indicate difficulty sustaining inhibitory control. Drifting later may indicate attention fatigue.

CLINICAL REFERENCE

Clinical Interpretation Scale

Accuracy

Range	Interpretation	Action
85–100%	Functional Skill established	Advance level or increase BPM by 4–8
70–84%	Emerging Moderate consistency	Continue; target 85% across 2 sessions
50–69%	Below threshold	Adjust supports; consider slower BPM
<50%	Well below threshold	Reduce BPM by 8–10 or drop one level

Timing Deviation (lower is better)

Range	Interpretation	Clinical Relevance
0–20ms	Excellent	Strong internal clock; ready for more complex tasks
21–40ms	Good	Typical for children in active development
41–70ms	Moderate difficulty	Likely affecting handwriting, bilateral tasks, reading rhythm
71–100ms	Significant difficulty	Document functional impact; direct intervention warranted

Range	Interpretation	Clinical Relevance
100ms+	Severe difficulty	Reduce demands; prioritize sensory feedback supports

Direction & Session Trend

Pattern	May Indicate
Consistently + (early)	Impulsivity, response inhibition — common in ADHD and anxiety
Consistently – (late)	Processing speed delay, motor initiation difficulty — common in DCD, TBI, fatigue
Stable across session	Consistent regulation throughout
Drifting early over time	Mounting impulsivity or inhibitory control difficulty
Drifting late over time	Attention fatigue or processing slowdown under demand

Key documentation point

Always report both accuracy AND deviation. Connect the deviation direction to a functional impact in daily occupations — handwriting, turn-taking, dressing, gait, reading rhythm.

DOCUMENTATION

SOAP Note Guide

When the session ends, fill in the **Session Observation** field on the results screen — then tap **Generate SOAP Note**. The note is pre-populated with all session data. Tap  Email to send to your saved address (set in  Settings → Therapist Email).

Session Observation field — fills the S section automatically

Type your clinical observation in the text box on the results screen — patient engagement, behavior, fatigue, comments made during the session. Whatever you enter populates the S section automatically. If left blank, the S section shows a standard fill-in prompt.

Section	Content
S	Populated from Session Observation field — or blank prompt if not filled in
O	Accuracy %, deviation with direction, session trend, level, BPM, modality, age category, hardware confirmed
A	Plain-language interpretation with functional connection prompts and progression status
P	Specific next-session BPM and level recommendation based on this session's data

Always edit before use

Draft only. The A section has fill-in prompts — review and personalize before pasting into your EMR or submitting for billing.

Insurance documentation tip

Add the functional connection in the A section — how does this patient's timing deficit show up in handwriting, dressing, reading, turn-taking, or gait? The objective numbers are already there.

CUSTOMIZATION

Settings Reference

Setting	What It Does
Beat Sound	Single Beat · Double Beat · 1–2 Spoken (says "one" and "two" — good for young children)
Audio Feedback	Spatial audio on/off — left ear tone = tapping early, right ear tone = tapping late. Requires wired headphones.
Metronome	Toggles the beat click — turn off to test internal timing without auditory cue
Device Haptics	Phone or tablet vibration on each tap
XAC Pillow Rumble	Drives vibration cushion via XAC. Choose Correct tap (reward) or Timing error (error correction).

Setting	What It Does
Blind Training Mode	Fades visuals after 10s — trains audio/haptic timing only
Therapist View	Shows ms deviation and clinical notes in real time during session
Track Accuracy (Vocab)	Off = listen-only mode — patient says letters aloud, no tapping required
Therapist Email	Saved email address — pre-fills  Email button on every SOAP note
Session Setup Prompt	Hardware confirmation before each session
Speech (Cognitive)	Speaks letters, words, and numbers aloud on the beat
Haptic Intensity	Slider for device vibration strength (mobile only)

SETUP

Calibration

Eye Tracking Calibration (WebGazer)

Available in Smooth Pursuits 2 and Attention Hold sub-modes. Uses front camera to estimate where the patient's eyes are pointing. Screening-level only — not clinical-grade eye movement recording. Chrome desktop + WebGazer required.

Requirements: Chrome on Windows, Mac, or Android. Not supported on iPhone, iPad, or Safari.

How to calibrate: Enable toggle → tap Play → 9-point calibration grid → look at each dot and tap it → tap Start Session. Gaze crosshair appears on canvas during session.

Accuracy note

WebGazer accuracy is approximately ±60–80 pixels depending on device and lighting. Screening-level — enough to detect clear trailing or anticipation, but not precise enough to replace a behavioral optometry evaluation. SOAP note includes appropriate disclosure automatically.

Camera Positioning (Motion & Active Games)

Positioning guide appears in the setup overlay for each sub-mode. Start button enables automatically when required body landmarks are detected. Full-body modes wait until both ankles are visible.

Mode	Camera Shows	Distance
Clapping	Waist up — both hands visible	3–4 feet
Knee Taps / Midline	Waist to head, knees visible	3–5 feet
Marching	Full body — head to feet	4–6 feet
Jumps	Full body + clearance above head	5–7 feet
All Active Games	Full body	4–7 feet

Camera zoom tip

A zoom slider (1×–2×) appears in the camera setup screen. Slide right so the patient can stand closer. Pose detection remains accurate up to approximately 1.5×.

EQUIPMENT

Hardware Guide

EvoSensory works on any device with a browser — no accessories required. On a computer tap with **spacebar** or **Enter**. On a phone or tablet, tap anywhere on screen.

Wired headphones — required for spatial audio feedback

The spatial audio is the core clinical mechanism: left ear tone = tapping early, right ear tone = tapping late. Without wired headphones this feedback is not available. Bluetooth adds 100–200ms of lag that defeats the feedback loop entirely. Any wired earbuds work — including old phone earbuds.

Recommended Accessories

Vibrating Cushion — for added sensory feedback

Place behind the patient (against the seatback) or under them on the seat. In Settings → XAC Pillow Rumble, choose when it fires:

Correct tap (reward)

Vibrates on clean hits. Best for early learners and sensory-seeking patients.

Timing error (correction)

Vibrates when patient taps early or late. Pairs with spatial audio cue.

External Button / Switch — for foot taps

Any USB switch or jelly bean button works as foot tap input. Level 6 is designed for bilateral foot training. The app automatically sets a lower starting BPM for foot modality.

Xbox Adaptive Controller (XAC) — connects everything

To connect external buttons/switches *and* a vibrating cushion in one device, the XAC is the recommended hub. AAC switches plug into the 3.5mm back jacks — no adapter needed. Button 0 = LEFT tap, Button 1 = RIGHT tap. USB connection means no Bluetooth lag.

The XAC gives you simultaneous L/R switch input for bilateral training and drives the vibration cushion — all in one device with a single USB connection.

At a Glance

Setup	Cost	What It Adds
Any device + browser	\$0	Full timing training, all modules, SOAP notes. Tap with spacebar, Enter, or screen tap.
+ Wired headphones	~\$10	Required for spatial audio. Left ear tone = tapping early, right ear tone = tapping late.
+ Vibrating cushion	~\$30–80	Whole-body proprioceptive feedback — reward or error correction (your choice)
+ USB switch / jelly bean	~\$20–60	External button for hand or foot input; larger tap target

Setup	Cost	What It Adds
+ Xbox Adaptive Controller	~\$100	Connects switches + cushion together. Bilateral L/R input + cushion output via one USB.
Chrome on computer	—	Required for Motion, Active Games, and Oculomotor eye tracking

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